

Leased Physical Capability

A Fail-Closed Safety Architecture for Autonomous Machines

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Abstract: This paper considers the safety implications of granting persistent physical authority to autonomous controllers. We introduce *Leased Physical Capability* (LPC), in which hazardous capability is granted for bounded intervals and must be renewed by mechanisms external to the controller. The controller may plan and request actions, but continued access to force, speed, mobility, energy, and operating duration depends on a trusted actuation gate, local authorization, and non-programmable physical limits. Lease expiration invokes a verified minimum-risk transition rather than arbitrary de-energization. LPC combines leases, reference monitors, runtime assurance, and functional safety, with the final bound placed in the physical actuation path.

Index Terms: autonomous systems, robotics safety, runtime assurance, functional safety, reference monitor, capability control

I. INTRODUCTION

Machine safety commonly relies on guards, monitored states, separation, emergency stopping, and inherently safer design. Runtime assurance (RTA) adds a monitor and recovery mechanism around a complex or insufficiently verified controller; ASTM F3269 provides such a framework for aircraft functions containing complex behavior [4]. These techniques, however, do not directly address persistent physical authority once force, mobility, energy, and operating duration have already been granted.

The issue becomes more significant as autonomous machines become smaller, faster, and less observable. Safety may then depend on the continuing correctness of software, sensors, communications, and recovery logic. Existing standards only partly cover this space. ISO 13482:2014 excludes flying robots and robots above 20 km/h, while ISO 10218-1:2025 excludes airborne robots, consumer products, and public-access service robots [5], [6]. ISO/FDIS 13482 broadens the successor standard to service robots [6]. This paper therefore considers a different design objective: instead of coupling intelligence to continuing physical authority, bound the authority independently of the controller.

II. LEASED AUTHORITY

Distributed systems use leases as time-limited grants that expire if renewal fails [1]. LPC applies the same principle to physical capability. Conventional authorization assumes

authorization remains valid until revoked,

whereas LPC assumes

authorization disappears unless renewed.

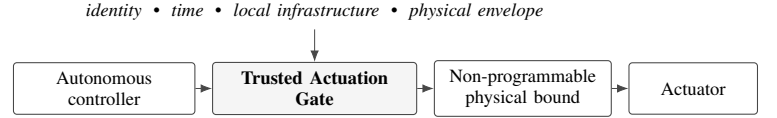


Fig. 1. LPC places an independently authorized gate and a non-programmable physical bound beneath the autonomous controller.

Thus, *intelligence is not authority*. Planning capability does not itself confer physical capability. Authority is external to the controller, specific to a function, bounded in magnitude and duration, temporary, revocable, and constrained by mechanisms that the controller cannot reprogram.

Let the authorization for elevated capability at time t be

$$P(t) = I(t) \wedge T(t) \wedge Z(t) \wedge E(t) \wedge H(t), \quad (1)$$

where I is independently authenticated identity, T an unexpired time lease, Z a fresh grant from trusted local infrastructure, E compliance with independently imposed physical limits, and H integrity of the trusted actuation path. If any term becomes false, renewal ceases. The controller may request renewal, but cannot grant, extend, or fabricate its own authorization.

III. TRUSTED ACTUATION GATE

Every safety-relevant actuator is reached through a small *Trusted Actuation Gate* (TAG), as shown in Fig. 1. The construction is analogous to Anderson's reference monitor concept, implemented by a reference validation mechanism that is tamper-resistant, always invoked, and sufficiently small for analysis and test [2]. The TAG accepts commands only while the applicable lease remains valid and independently established physical limits are satisfied.

RTA and control-barrier methods may detect or filter unsafe trajectories [3], [4]. LPC addresses a separate question: whether the controller retains the physical authority required to execute the trajectory. RTA constrains runtime behavior; LPC constrains the persistence and magnitude of authority available to the controller.

IV. LEASE EXPIRATION AND MINIMUM-RISK TRANSITION

Fail-closed operation must not be interpreted as unconditional power removal. An autonomous heavy vehicle that loses communications on a railway crossing may require traction to leave the crossing; immediate de-energization can therefore increase risk. Each leased capability must instead define a verified *minimum-risk transition* (MRT). Expiration terminates

authority to continue the mission while preserving only the bounded actuation required to reach a certified minimum-risk state. This is related to the minimal-risk-condition concept used in automated-driving guidance [8]. Starter-interrupt devices illustrate the same distinction: denying the next start differs materially from forcing an engine to stop while a vehicle is moving [9]. Revocation should therefore occur at the earliest state at which revocation is itself safe.

V. NON-PROGRAMMABLE PHYSICAL BOUNDS

The essential requirement is that *the maximum hazardous capability be bounded by something the controller cannot re-program*. A physical fuse is one implementation; others include hardwired drive limits, mechanically constrained transmissions, independently configured safety hardware, fixed energy-delivery limits, and deliberately limited energy storage.

A small battery, fuel store, or capacitor bank can bound persistence and cumulative work before return to trusted infrastructure, while a fixed energy-delivery rate can bound sustained power. Energy storage alone does not bound peak force or instantaneous release, so separate hard limits may still be required. The relevant test is whether complete compromise of the controller can increase the certified physical envelope. If it can, the bound is not independent.

VI. LOCAL PHYSICAL AUTHORIZATION

LPC need not treat location as a primitive. The TAG need not obtain a coordinate from the controller and compare it with a virtual boundary. Elevated capability may instead depend on authorization emitted by stationary trusted infrastructure whose physical reach defines where renewal is possible:

$$Z = \begin{array}{l} \text{authorization obtained from} \\ \text{locally unavoidable infrastructure.} \end{array} \quad (2)$$

A machine exercising elevated capability therefore remains physically dependent on infrastructure fixed to the authorized environment. Leaving its physical reach removes the condition required for renewal. The system need not prove an arbitrary coordinate; it need only establish continued coupling to the local condition on which authorization depends.

Cryptography authenticates the issuer of a grant but does not establish locality. The authorization channel should therefore exploit physical properties such as bounded propagation time, short-range coupling, fixed optical paths, or conductive and inductive interfaces. Received signal strength alone is inadequate against relay or amplification. Remote ID demonstrates the value of broadcast identity and location, but because the aircraft itself supplies that information it is not an independent local authorization mechanism [7].

VII. CAPABILITY AND CONSPICUOUSNESS

Miniaturization can reduce observability and warning time without eliminating useful computation or mobility. Permitted capability should therefore be considered jointly with detectability, attribution, stopping capability, and containment. As machines become smaller, faster, less observable, or more

numerous, independently available energy, hazardous momentum, persistence, and unrestricted authority should decrease unless stronger containment is demonstrated. Greater hazardous capability should, conversely, require greater conspicuousness and greater dependence on external infrastructure. This is not a minimum-size rule; the relevant relation is between physical capability and the ability to observe, attribute, interrupt, and contain it.

VIII. DISCUSSION

LPC changes the design objective from attempting to establish that arbitrary intelligence is safe to ensuring that arbitrary intelligence remains physically bounded:

$$\begin{array}{c} \text{make arbitrary intelligence safe} \\ \downarrow \\ \text{make arbitrary intelligence physically bounded.} \end{array}$$

Dangerous authority should decay toward a safe state unless an independent mechanism continuously renews it. Such authority should be external, specific, bounded, temporary, revocable, and physically constrained.

A related implication concerns connectivity. Safety-critical actuators, utilities, transport interlocks, and refuge systems should remain capable of safe operation without cloud services, general-purpose networks, remote administrative paths, or autonomous planners. For such systems, connection should be treated as another leased capability rather than as the default condition.

The immediate research problems are to specify a minimal TAG; derive lease duration and MRTs from system dynamics; construct relay-resistant local authorization; characterize physical bounds in force, power, energy, and persistence; and define capability-conspicuousness metrics. Lease-channel denial must not introduce a new hazard. These questions admit direct experimental treatment and provide a path from the proposed architecture to testable safety requirements.

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